Securing AI and Advanced Topics

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
Fraud Prevention with Cloud AI Solutions

Module 3
Introduction to Generative Adversarial Attacks (GANs)

Module 4
GANs and Adversarial Attacks

Module 5
Reinforcement Learning

Module 6
Evaluating AI Models and Performance

Feature Engineering and Model Optimization Techniques

- Challenges with Using AI for Cybersecurity
  Video • 8 min
- Reading References
  Reading • 10 min
- Feature Engineering and Model Optimization Techniques
  Practice Assignment • Grade: 100%

Evaluating AI Models and Performance Metrics

- Evaluating AI Models
  Video • 7 min
- Reading References
  Reading • 10 min
- Evaluating AI Models and Performance Metrics
  Practice Assignment • Grade: 100%

Module-end Assessments

- **Self-Reflective Reading: Feature Engineering in Cybersecurity Applications**
  Reading • 45 min
- Evaluating AI Models and Performance
  Graded Assignment • 1h

Self-Reflective Reading: Feature Engineering in Cybersecurity Applications

0:05 / 3:20 2x

Introduction

In this module, you explore various mechanisms for evaluating AI models, focusing on the intricacies of feature engineering, data manipulation, and performance assessment. Understanding how to manipulate raw data and engineer features is critical for improving model performance, especially in domains like cybersecurity. This module guides you through comparisons of machine learning models applied to different cybersecurity problems, and will encourage you to dive deeper into cross-validation techniques as an alternative to traditional dataset splitting.

Additionally, you will conduct a literature review to assess the latest advancements in feature engineering and performance metrics, comparing these methods to those discussed in your lectures.

Tasks to Complete:

1. **Cite Examples of Feature Engineering**: Choose a machine learning example and discuss the feature engineering and raw data manipulation techniques used, using terminology from this module's lectures.
2. **Compare ML Models for Cybersecurity Problems**: Select the same ML model applied to different cybersecurity issues. Compare the performance using this module's performance measurement terminology.
3. **Retrain Classifiers with Cross-Validation**: Instead of splitting the dataset into training and testing subsets, apply cross-validation and observe whether performance improves.
4. **Conduct a Literature Review**: Investigate current feature engineering techniques and performance metrics in AI research and compare these with the methods covered in the lecture.

Steps:

- **Step 1**: Identify and cite examples of machine learning models with raw data manipulation and feature engineering approaches. Discuss these using terminology from your lectures.
- **Step 2**: Review and compare performance measures of ML models used for different cybersecurity applications. Use appropriate performance metrics like precision, recall, and F1-score.
- **Step 3**: Apply cross-validation techniques to retrain classifiers and evaluate whether cross-validation yields better performance metrics than the traditional train-test split.
- **Step 4**: Conduct a brief literature review on feature engineering and performance metrics in AI. Compare these findings with the approaches covered in your lectures.

Reflective Questions:

1. **Feature Engineering Reflection**: How did the chosen feature engineering mechanisms influence model performance? Could alternative techniques have improved the outcome?
2. **Comparing Models**: What were the key differences in the model's performance across different cybersecurity problems? Did specific metrics (e.g., precision, recall) vary significantly? Why?
3. **Cross-Validation Impact**: How did cross-validation impact the model's performance compared to traditional dataset splitting? Were there any improvements in consistency or generalizability?
4. **State-of-the-Art Comparison**: After reviewing the literature, how do the latest feature engineering and performance metrics compare to what was covered in the lecture? Are there emerging trends or technologies that could enhance current practices?

This framework should help guide your evaluation process while offering reflective insights into the techniques and methodologies discussed in this module.

Go to next item ✓ Completed

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